PS-05 Groundnut Leaf Disease Detection

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Domain: Agriculture & Food

Classification of field-collected groundnut leaf images using EfficientNet-B0

Primary metric	Test result	Secondary metrics
Macro F1	0.9705	Accuracy 0.9755 | Balanced Accuracy 0.9704

1. Problem and Objective

Groundnut is an important rain-fed crop in South India, and leaf diseases can reduce plant health and yield if symptoms are not recognized early. The objective was to build an image classifier that assigns a groundnut leaf to one of five target classes: Early Leaf Spot, Late Leaf Spot, Rust, Nutrition Deficiency, or Healthy. Because the class distribution is unequal, overall accuracy alone can hide weak minority-class performance. Therefore, Macro F1 was selected as the primary success metric, with balanced accuracy and per-class precision/recall/F1 used as supporting measures.

2. Dataset and Pre-processing

The dataset was the Groundnut plant leaf image dataset from Mendeley Data (Version 3): https://data.mendeley.com/datasets/22p2vcbxfk/3. The extracted release contained six source folders; to match the five-class PS-05 specification, early_rust_1 and rust_1 were mapped to the single target class Rust. This produced 7,910 supplied training images and 2,451 supplied test images. The supplied test set was kept untouched, while 20% of the supplied training portion was stratified into validation data, giving 6,328 training, 1,582 validation, and 2,451 test images. Images were resized to 224×224 pixels and normalized using ImageNet mean and standard deviation. Training-only augmentation used random horizontal flipping, ±10° rotation, and mild brightness/contrast jitter. Class-weighted loss was used to compensate for unequal class frequencies.

Split	Images
Training	6,328
Validation	1,582
Test	2,451
Total	10,361

Target class	Training	Validation	Test
Early Leaf Spot	1,057	265	409
Late Leaf Spot	1,193	298	405
Rust	1,904	476	818
Nutrition Deficiency	1,004	251	410
Healthy	1,170	292	409

3. Methodology

Transfer learning was performed with an ImageNet-pretrained EfficientNet-B0 backbone. Its final classifier was replaced by a five-output linear layer (1,280 input features → 5 classes), and the full network was fine-tuned rather than freezing the backbone. Weighted cross-entropy was used so that errors on less frequent classes contributed more strongly to the objective. AdamW optimized all model parameters with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4}. A ReduceLROnPlateau scheduler monitored validation Macro F1, halving the learning rate after two non-improving epochs. Training ran for 10 epochs with batch size 32. Formal early stopping was not used; instead, the checkpoint with the highest validation Macro F1 was saved and restored for final testing.

Hyperparameter	**Value**
Backbone	EfficientNet-B0, ImageNet pretrained
Fine-tuning	All layers
Input size	224 × 224 RGB
Loss	Class-weighted cross-entropy
Optimizer	AdamW
Learning rate	0.0001
Weight decay	0.0001
Batch size	32
Epochs	10
Scheduler	ReduceLROnPlateau, factor 0.5, patience 2
Model selection	Best validation Macro F1 checkpoint

4. Results

The best validation Macro F1 was 0.9971. On the held-out test set of 2,451 images, the selected model achieved 97.55% accuracy, 97.04% balanced accuracy, and a Macro F1 of 0.9705. Training time was not recorded in the notebook, so it is reported as not available rather than estimated.

Model	**Test Accuracy**	**Balanced Accuracy**	**Macro F1**	**Training time**
EfficientNet-B0	0.9755	0.9704	0.9705	Not recorded

Class	**Precision**	**Recall**	**F1**	**Support**
Early Leaf Spot	0.8975	0.9633	0.9292	409
Late Leaf Spot	1.0000	0.8889	0.9412	405
Rust	1.0000	1.0000	1.0000	818
Nutrition Deficiency	1.0000	1.0000	1.0000	410
Healthy	0.9646	1.0000	0.9820	409



Figure 1. Training and validation loss across 10 epochs.

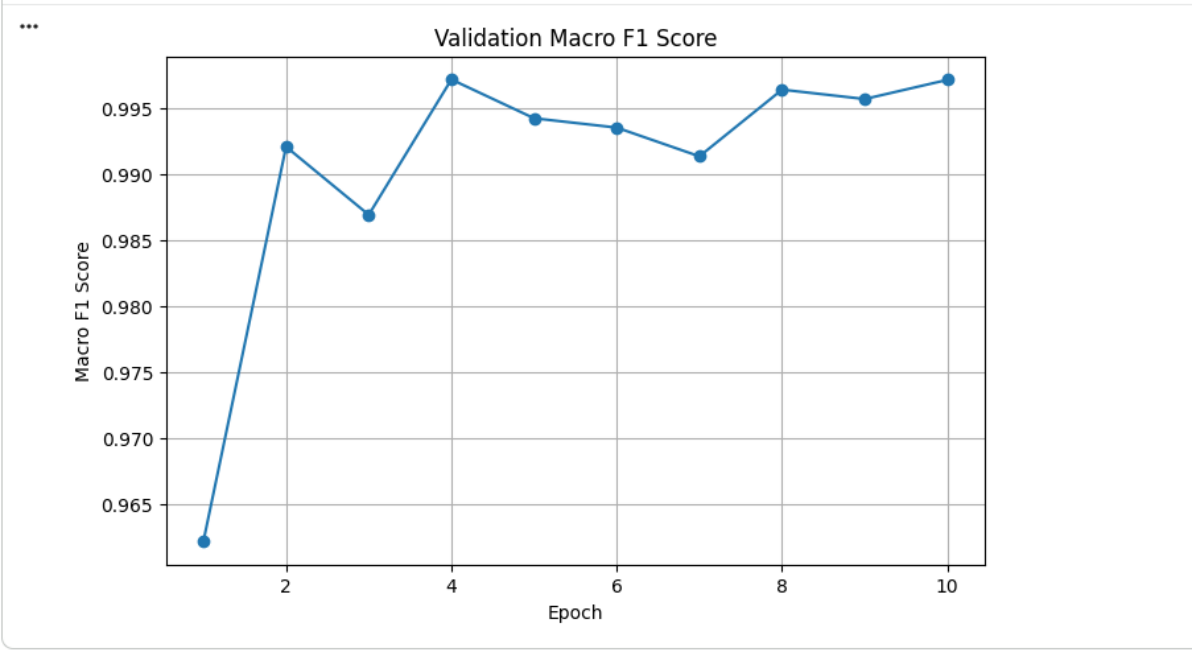


Figure 2. Validation Macro F1 across 10 epochs; the best value was 0.9971.

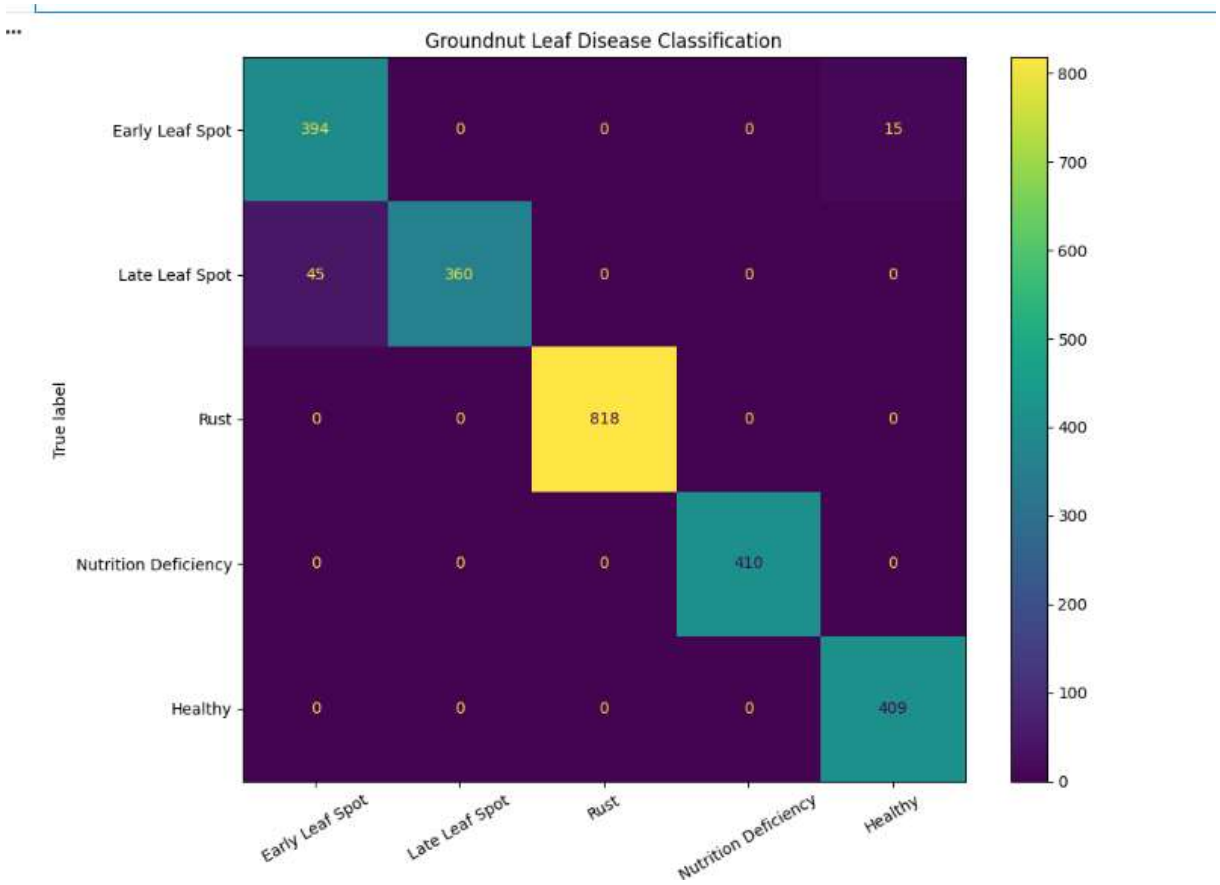


Figure 3. Confusion matrix on the 2,451-image test set.

5. Analysis and Discussion

The model performed strongly overall, but the remaining errors were concentrated in visually similar leaf-spot classes. The confusion matrix shows that 45 Late Leaf Spot images were predicted as Early Leaf Spot, while 15 Early Leaf Spot images were predicted as Healthy. This explains why Late Leaf Spot had the lowest recall (0.8889) and Early Leaf Spot had the lowest precision (0.8975). In contrast, Rust and Nutrition Deficiency achieved perfect precision, recall, and F1 on this test split, and Healthy achieved an F1 of 0.9820. Training and validation loss both decreased to low values, while validation accuracy and Macro F1 remained close to training performance, so the learning curves do not show strong conventional overfitting. A sampled prediction also illustrates the main failure mode: one Late Leaf Spot image was classified as Early Leaf Spot with 94.11% confidence. The imbalance challenge was addressed through class-weighted cross-entropy and evaluation with Macro F1/balanced accuracy rather than relying only on overall accuracy.



Figure 4. Sample test predictions (part 1), showing actual class, predicted class, and confidence.

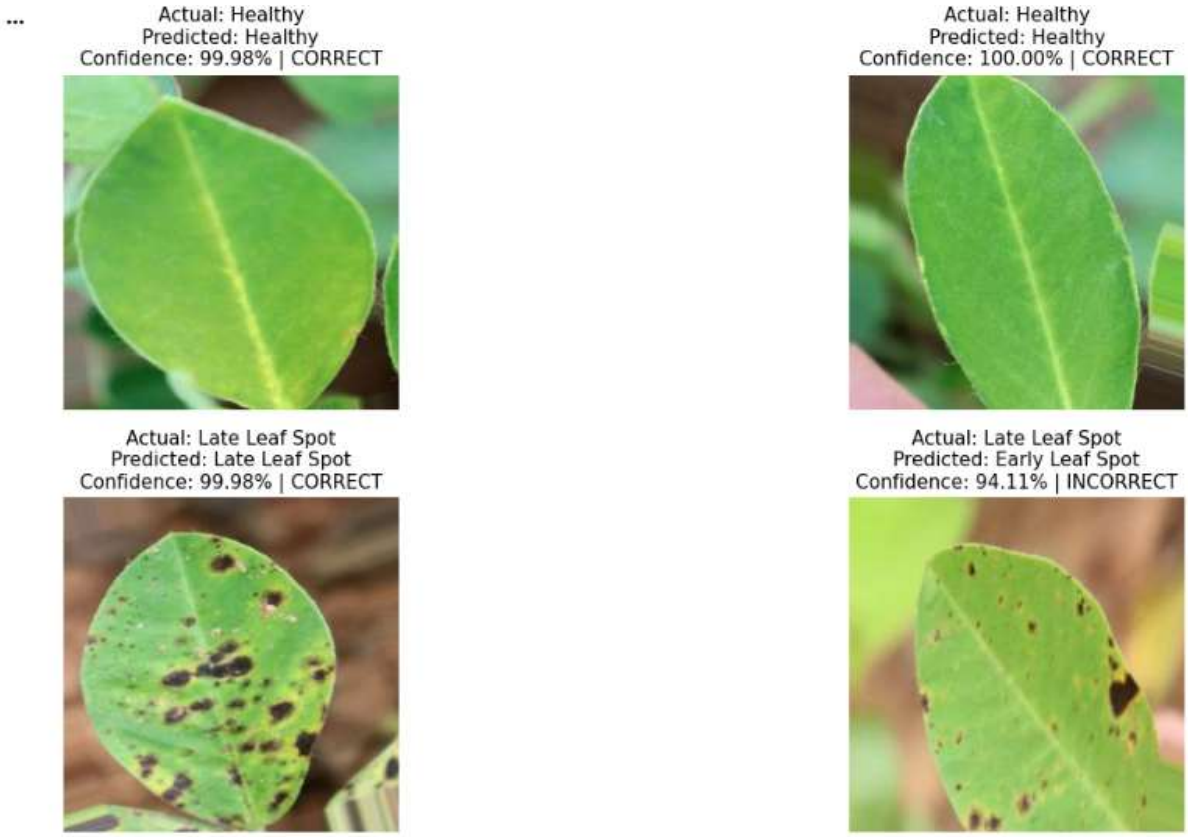


Figure 5. Example qualitative predictions, including the Late Leaf Spot misclassification.

6. Real-world Relevance, Limitations and Ethics

A model of this type could support farmers, extension workers, or mobile crop-monitoring tools by providing rapid screening of photographed groundnut leaves. It should not yet be treated as a stand-alone diagnostic system: performance must be verified on independent farms, seasons, devices, lighting conditions, cultivars, and disease severities. The very high scores on some classes also warrant external validation, especially because the downloaded release contains derived-looking image names and a six-folder source structure that was mapped to five targets. Misclassification could lead to inappropriate crop-management decisions, so predictions should be presented with

uncertainty and expert confirmation where consequences are significant. Farm images should also be handled with appropriate consent and privacy safeguards if they include people, location metadata, or identifiable property.

7. Conclusion and Future Work

The fine-tuned EfficientNet-B0 achieved a test Macro F1 of 0.9705 and 97.55% accuracy for the five-class groundnut leaf classification task, with most remaining confusion occurring between Early and Late Leaf Spot. The next improvement should be evaluation on a strictly independent field dataset and targeted hard-example training for the two leaf-spot classes, while verifying that augmented or related source images cannot leak across data splits.